

Partial replacement of Cement with Agriculture Waste Ash

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ABSTRACT: The construction industry is continuously searching for sustainable and eco-friendly materials to reduce environmental pollution and the excessive use of natural resources. Cement production releases a significant amount of carbon dioxide, which contributes to global warming. One possible solution is the partial replacement of cement with agricultural waste ash, which utilizes waste materials and reduces the demand for cement. Agricultural wastes such as Rice Husk Ash, Bagasse Ash, and Wheat Straw Ash contain a high amount of silica that can react with cement and improve the properties of concrete.

In this study, agricultural waste ash is used as a partial replacement for cement in different proportions to evaluate its effect on the strength and durability of concrete. The aim is to determine the optimum percentage of ash that can replace cement without significantly reducing the strength of concrete. The use of agricultural waste ash not only helps in waste management but also reduces construction costs and environmental impact. The results indicate that a suitable percentage of ash can be effectively used in concrete production while maintaining satisfactory strength and performance.

Keywords: Agricultural Waste Ash, Cement Replacement, Sustainable Construction, Concrete Strength, Waste Utilization, Eco-friendly Materials

INTRODUCTION:

The construction sector is one of the largest consumers of natural resources and a major contributor to global carbon emissions. Primarily due to the extensive use of Ordinary Portland Cement (OPC) in concrete production. The manufacturing of cement is highly intensive and release a considerable amount carbon dioxide (CO₂) making it a significant sources of greenhouse gas emission. In recent years, growing environmental concern and the depletion of natural resources have encouraged researches to explore to sustainable alternatives to conventional cementitious materials. Among the various approaches proposed, the partial replacement of cement with agriculture waste ash has emerged as a promising strategy for reducing the environmental burden associated with concrete production.

Agriculture waste ash, derived from the controlled combustion of agro-residues such as rice husk, sugarcane bagasse and coconut shell, has gained increasing attention as a supplementary cementitious material in sustainable concrete technology. These ashes are generally rich in silica and other reactive oxides, which contribute to pozzolanic reaction in cementitious systems. When incorporated in suitable proportions, agricultural waste ash reacts with calcium hydroxide released during cement hydration and forms additional cementitious compounds, thereby improving the microstructure of concrete. This secondary reaction enhances the bonding characteristics of the cement matrix and contributes to improved long-term performance.

Several studies have reported that the inclusion of agriculture waste ash in concrete can improve selected engineering properties while simultaneously reducing cement consumption, Rice Husk Ash (RHA) in particular has been widely recognized for its high amorphous silica content and strong pozzolanic reactivity,

which can enhance compressive strength and durability when used at controlled replacement levels. Similarly, Sugarcane Bagasse Ash (SCBA) has shown potential in improving resistance to chemical attack and refining pore structure due to its silica-alumina composition. Coconut Shell Ash (CSA), though comparatively less reactive, has also demonstrated beneficial effects in improving thermal behavior and reducing the overall environmental footprint of concrete. The incorporation of these waste-derived ashes not only improves material utilization but also promotes the conversion of agriculture residue into value-added construction resources.

The environmental significance of utilizing agricultural waste ash in concrete extends beyond cement reduction. In many agriculture regions, biomass residues are commonly disposed of through open-field burning or uncontrolled dumping, both of which release serious environmental concerns. Open burning releases particulate matter and harmful gases into the atmosphere, while improper dumping contributes to land degradation and waste accumulation. Converting these residues into ash for cement replacement offers a practical and environmentally responsible waste management solution. This approach supports waste valorization by transforming low-value agricultural by-product into functional construction materials, thereby contributing to cleaner production practices and circular resource utilization.

From an engineering perspective, the use of agricultural waste ash in concrete has shown promising results in terms of strength development, workability and durability performance. The effectiveness of these ashes, however, largely depends on their chemical composition, fineness, and replacement ratio. Lower replacement levels generally improve mechanical performance due to enhanced pozzolanic activity and particle packing, whereas excessive replacement may reduce strength because of insufficient cementitious content. Therefore, identifying an optimum replacement level is essential to achieve a balance between structural performance and environmental sustainability.

Despite the growing interest in agro-waste-based cementitious systems, a comprehensive understanding of their performance under varying replacement levels remains necessary for broader practical application. Most previous studies have focused on individual waste materials or isolated performance indicators, while limited attention has been given to comparative evaluation under a unified experimental framework. In this context, the present study investigates the feasibility of partially replacing the cement with agriculture waste ash in concrete and evaluates its influence on workability, compressive strength, water absorption, and density. Different replacement levels were considered to determine the optimum dosage that provides improved material performance while reducing environmental impact. By integrating engineering performance with sustainability considerations, this study aims to contribute to the development of eco-efficient and low-carbon construction materials for future infrastructure applications.

1. MATERIALS AND METHODOLOGY:

1.1. MATERIALS

1.1.2. Cement : Cement is an important binding material. It acts as the main ingredient in the mixture, which binds all components together. It also contributes to strength and durability. Ordinary Portland Cement (OPC) helps in achieving comparatively higher performance.

1.1.3. Crushed Aggregate: Crushed stones of size 10-20 mm are used. These aggregates are selected based on their strength and proper shape, which provide adequate load-carrying capacity and structural stability.

1.1.4. Water: Water plays a crucial role in both mixing and curing process. It ensures proper hydration and is responsible for achieving the desired strength and quality of concrete.

1.1.5. Agricultural waste: Agricultural waste are used as partial replacement in cement to make the concrete more sustainable these materials improves certain properties and help in proper environmental conservation example- Rice husk, sugarcane bagasse, coconut shell ash agriculture wastage are rich in silica and alumina, which provide pozzolanic properties these materials improve hydration reaction, enhance strength and durability, reduce environmental impact are locally available and economical supports sustainable development.

2.2 METHODOLOGY:

Table:1 Proportion of Replaced materials

Material	Purpose in Concrete mix	Replacement Level
Ordinary Portland Cement (OPC)	Primary binding material	100%,95%,90%,85%
Agriculture Waste Ash	Partial replacement of cement	0%,5%,10%,15%
River Sand	Fine aggregate	Constant proportion
Coarse Aggregate (10-20mm)	Structural strength and bulk volume	Constant proportion
water	Hydration and workability	As Required
water-Binder Ratio	Controls consistency and strength	0.45

Table:2 IS Code Reference for Material standards

Materials	Relevant Standard
Ordinary Portland cement (OPC)	IS 12269:2013
Fine Aggregate (River Sand)	IS 383:2016
Coarse Aggregate	IS 383:2016
Water for Mixing and Curing `	IS 456:2016
Compressive Strength Test	IS 516 (Part 1) :2021
Slump Test	IS 1199:2018
Agriculture Waste Ash	Evaluated as supplementary cementitious material

2.2.1 Collection and preparation of Agriculture waste: Agriculture waste materials were collected from local sources and processed before use. Rice husk, sugarcane bagasse and coconut shells were collected from nearby agriculture and domestic waste sources. These materials were cleaned properly to remove impurity such as dust, soil, and unwanted particles. The cleaned materials were then dried and burned under controlled conditions to obtain ash. The ash obtained was sieved through a fine sieve to remove larger particles and achieving uniform fineness. The prepared ash was stored in dry conditioned for further use in concrete mixing.

2.2.2 Mixing of Concrete: Concrete mixing was carried out carefully to ensure uniform distribution of materials. First cement, fine aggregate, coarse aggregate, and agriculture waste ash were mixed in dry conditioned until a uniform color was obtained. After dry mixing, water was added gradually in the required quantity. The materials were then mixed thoroughly to obtain a homogeneous concrete mix. Proper care was taken to avoid segregation and ensure uniform consistency.

2.2.3 Casting of Specimens: After mixing, the concrete was placed into moulds for specimen preparation. Cube moulds were cleaned and oiled before use to prevent sticking. Fresh concrete was poured into the

moulds in layers. Each layer was compacted properly using tamping rods or vibration to remove entrapped air. The top surface was leveled and finished smoothly. The specimens were left undisturbed for 24 hours.

2.2.4 Curing of Specimens: After 24 hours, the specimens were removed from the moulds and cured properly. The cast specimens were demoulded carefully. They were then immersed in clean water for curing. Curing was carried out for different periods such as 7 days, 14 days, and 28 days. Proper curing ensured complete hydration and strength developed.

2.2.5 Testing of Concrete: After curing, the concrete specimens were tested to evaluate their performance. The specimens were tested for compressive strength using a compressive Testing machine (CTM). The load was applied gradually until failure occurred. The maximum load carried by the specimen was recorded. Compressive strength was calculated and compared for different mixes.

2.2.6 Analysis of Results: The test results were analyzed to study the effect of agricultural waste ash on concrete properties. Strength values of conventional concrete and modified concrete were compared. The optimum percentage of agricultural waste replacement was identified. The performance was evaluated in terms of strength, workability and sustainability. Final conclusions were drawn based on experimental observations.

3. Environment safety Assessment

Environment safety assessment refers to the process of evaluating how human activities affect the environment and identifying measures to reduce environment harm. In this context, the assessment focuses on the environmental impact of cement production.

Cement production is considered harmful to environment because it releases large quantities of hazardous gases and pollutants into the atmosphere. These emissions contribute significantly to environment degradation and air pollution.

One of the major concerns associated with cement manufacturing is the high level of carbon dioxide emissions. Carbon dioxide is a dangerous greenhouse gas that contributes to global warming and climate change. Therefore, reducing carbon dioxide emission has become a critical environmental objective.

One effective solution is the partial replacement of cement with agricultural waste ash. This approach can significantly reduce carbon dioxide emission while also promoting the sustainable use of waste materials. Agriculture waste material such as rice husk coconut ash can be utilized as partial substitutes for cement in construction materials.

This approach not only reduces the environmental impact of cement production but also supports sustainable construction practices by converting agricultural waste into valuable construction materials. Reduction in Greenhouse Gas Emission through clinker production

Cement manufacturing is one of the major contributors to harmful emissions, particularly due to the large-scale release of carbon dioxide. Most of these emissions are generated during clinker production, which is one of the most energy intensive stages in cement manufacturing. To address this issue, the use of agriculture waste ash as a partial replacement of cement can help reduce cement demand and, consequently, lower overall carbon emissions.

This approach has gained increasing attention in research because it promotes the efficient use of natural resources while also helping to mitigate the adverse effect of climate change.

4. Sustainable Waste Valorization

Agriculture waste is often disposed of by open burning or dumping in landfills. These practices release harmful gases into the atmosphere, causing serious environmental pollution and long term ecological damage. Such uncontrolled disposal methods can become highly hazardous and contribute significantly to air pollution.

The utilization of agriculture waste in cement blocks or concrete offers a sustainable and environmentally friendly waste management solution. This “green method” of waste disposal not only reduces environmental pollution but also adds value to waste material by converting them into useful construction resources.

According to established literature in the field of sustainable materials, the conversion of agro-waste into construction materials significantly improves the efficiency of waste recycling process while promoting sustainable development in the construction industry.

5. Focusus on Sustainable Sourcing

Cement production relies heavily on natural raw materials such as limestone and clay. The use of waste ash as a partial replacement for these conventional material helps reduces dependence on limited natural resources. This approach lowers the pressure on non-renewable geological reserves and minimizes the environmental burden associated with raw material extraction.

A significant reduction in the consumption of non-renewable geological resources can be achieved through the incorporation of biomass ashes. This transition towards sustainable sourcing insures that the environmental impact of raw material extraction is minimized while maintaining the structural integrity of the concrete matrix.

6. Conclusion

1.The investigation confirms that agricultural waste ash, particularly rice husk ash(RHA), possesses significant pozzolanic properties, making it a viable and sustainable alternative to ordinary Portland cement(OPC). Its use a partial cement offer both environmental and engineering benefits.

2.It was observes that the mechanical strength of concrete improves at a replacement level of up to 10%. Beyond this threshold, a slight decline in compressive strength was noted due to the dilution of the cementitious matrix. Therefore, an optimum replacement level of 10% is considered most effective for achieving both strength and sustainability.

3.The effective utilization of agricultural waste help address major environmental challenges associated with open-field burning and landfill disposal. This practices significantly reduces the carbon footprint linked to clinker production and contributes to a cleaner and more sustainable environment.

4.The patial replacement of cement with locally available agricultural by products offers a cost-effective alternative for construction materials. This approach not only reduces material costs but also provides an economic solution for low-co st housing and sustainable infrastructure development.

REFERENCES

1. Abdullah, M. M. A. B., Hussin, K., & Zakaria, F. (2011). Potential of rice husk ash as a cement replacement material. *Journal of Materials Science and Engineering*, 5(2), 123–128.
2. ACI Committee 232 (2013). Use of natural pozzolans in concrete. *ACI Materials Journal*, American Concrete Institute.
3. Bui, D. D., Hu, J., & Stroeven, P. (2005). Particle size effect on the pozzolanic properties of rice husk ash. *Cement and Concrete Research*, 35(5), 912–920.
4. Cordeiro, G. C., Toledo Filho, R. D., & Fairbairn, E. M. (2009). Characteristics of sugarcane bagasse ash and its effect on mechanical properties of concrete. *Construction and Building Materials*, 23(2), 100–107.
5. Deepa, B., & Praveen, M. (2020). Experimental study on partial replacement of cement with agricultural waste ashes. *International Journal of Engineering Research & Technology (IJERT)*, 9(6), 45–50.
6. IS 10262:2019. Concrete Mix Proportioning – Guidelines. Bureau of Indian Standards, New Delhi.
7. IS 456:2000. Plain and Reinforced Concrete – Code of Practice. Bureau of Indian Standards, New Delhi.
8. IS 2386 (Part I–VIII):1963. Methods of Test for Aggregates for Concrete. Bureau of Indian Standards, New Delhi.

9. IS 516:1959 (Reaffirmed 2004). Methods of Tests for Strength of Concrete. Bureau of Indian Standards, New Delhi.
10. Jagadeesh, P., & Kumar, R. (2019). Use of sugarcane bagasse ash as a supplementary cementitious material in concrete. *International Journal of Civil Engineering and Technology (IJCIET)*, 10(4), 112–121.
11. Mehta, P. K., & Monteiro, P. J. (2014). *Concrete: Microstructure, Properties, and Materials*. McGraw-Hill Education.
12. Nair, D. G., Fraaij, A., & Klaassen, A. (2006). A study on the reaction mechanism of rice husk ash in cementitious systems. *Cement and Concrete Research*, 36(6), 1053–1059.
13. Nehdi, M., Duquette, J., & Damatty, A. (2003). Performance of rice husk ash concrete under different curing conditions. *Cement and Concrete Research*, 33(6), 1019–1025.
14. Patel, K., & Shah, D. (2021). Experimental investigation on concrete with partial replacement of cement by RHA & SCBA. *International Research Journal of Engineering and Technology (IRJET)*, 8(7), 980–986.
15. Siddique, R. (2008). *Waste Materials and By-Products in Concrete*. Springer-Verlag, Berlin.